Roboteam

Daily Maintenance Procedure - Bison Robotic Vehicle

Pre-Operation and Post-Operation Maintenance

1. General

This procedure is intended to ensure the safety, reliability and long service life of the vehicle.

- **Pre-Operation -** At the start of every working day and before every use of the vehicle.
- **Post-Operation -** At the end of every working day and before night operations.
- All checks are performed on level ground. Mark X in the "OK" or "Not OK" column on each line, and record any faults in Section 5.
- Any deviation or fault must be reported immediately to the Roboteam maintenance team and fixed before the vehicle is used again. If in doubt - stop and consult.

2. Pre-Operation Maintenance

#	Check	How to Check / OK When	OK	Not OK	Remarks
A. Preparation					
1	Platform power-up	Platform is on and the software has loaded.			
2	Initial preparation	Vehicle on level ground, engine off.			
B. Engine Compartment - Cold Check (engine off, power on)					
3	General inspection	No damage, loose parts or foreign objects.			
4	Brakes	No tears, cracks, oil or dirt.			
5	Leaks	No brake fluid, gear oil, fuel or engine oil leaks.			
6	Engine oil level	On the dipstick, within the hatched range.			
7	Gear oil level	Sight glass half full. Remove the top cover to refill.			
8	Fuel level	Sufficient for the planned activity.			
9	Batteries 12V x2	Nuts tight, terminals clean with no corrosion.			
10	Drive belt	No tears, wear or cracks.			
11	Alternator belt	No tears, wear or cracks.			
C. Around and On the Vehicle					
12	Underbody and axles	No cracks or breaks in the underbody, axles not bent.			
13	Towing eyes	Towing eyes intact.			
14	EMO buttons	All 3 buttons present and working - test by pressing.			
15	Tire pressure	Front 5, middle 7, rear 6 (PSI).			
16	Tire condition	No tears or wear, valve intact.			
17	Wheel nuts	In place and tightened.			
18	Track	Rubber intact, teeth secured, no abnormal wear.			
19	Cameras	4 surround (day and night) and mast - working and secured.			
20	Communication antennas	RF and cellular antennas present and secured.			
21	Bolts	Vehicle and weapon system bolts secured.			

#	Check	How to Check / OK When	OK	Not OK	Remarks
D. Hot Check (after 20 minutes of operation)					
22	Oils and leaks	Recheck engine and gear oil. No new leaks.			

3. Post-Operation Maintenance

#	Check	How to Check / OK When	OK	Not OK	Remarks
A. Before Engine Shutdown					
1	Oil spray	No oil or fluid spray in the engine compartment or on the cover.			
2	Brake drying	After water, mud or deep sand - apply the brakes lightly while driving slowly.			
B. Engine Compartment (after shutdown)					
3	General inspection	No damage, loose parts or foreign objects.			
4	Leaks	No leaks from hoses or the cylinder head.			
5	Drive / alternator belt	No tears or burn marks.			
6	Air filter	Clean off dust and dirt. Replace a worn filter (especially in sandy or dusty conditions).			
C. Around the Vehicle					
7	Tires	Wheels intact, tire pressure correct.			
8	Track	Rubber intact, teeth secured.			
D. Closing Up the Vehicle					
9	Cleaning	Remove mud, sand and dirt to prevent corrosion.			
10	Camera lenses	Clean all lenses.			
11	Refueling	95 octane gasoline as needed.			
12	Charging	Charge platforms as needed.			

Proprietary Information - Commercially Classified

4. Appendices

4.1. Technical Data

Item	Value
Fuel type	95 octane gasoline
Fuel tank capacity	32 liters
Engine oil	5W-40 synthetic
Engine oil capacity	2.3 liters
Gear oil (transmission)	75W-90 synthetic
Gear oil capacity	1.2 liters
Brake fluid	DOT 4
Chain oil (spray)	PTFE
Grease	Lithium
Tire pressure (PSI)	Front 5, middle 7, rear 6

4.2. Periodic Maintenance by Engine Hours

Daily checks are listed in the Pre-Operation and Post-Operation sections. Periodic maintenance is as follows:

Task	Interval
Chain lubrication	Every 25 engine hours
Grease	Every 25 engine hours
Engine oil and filter change	Every 100 engine hours
Gear oil change	Every 100 engine hours
Air filter	Daily cleaning in Post-Operation, thorough service every 100 engine hours, replace when worn

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5. Maintenance Log

5.1. Pre-Operation Maintenance

Date	Responsible	Faults	Remarks	Signature

5.2. Post-Operation Maintenance

Date	Responsible	Faults	Remarks	Signature

Date	Responsible	Faults	Remarks	Signature